



CLOUD COMPUTING IN **Healthcare**

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Cloud computing definition

- Cloud computing is a resource delivery and usage model, it means get resource (Hardware, software) via network. The network of providing resource is called 'Cloud'.



Cloud - Service & Deployment Models

Cloud Service Models

Infrastructure as a Service

For processing, storage, networks, and other computing infrastructure resources

Platform as a Service

To deploy applications developed using specified programming languages or frameworks and tools onto the cloud infrastructure

Software as a Service

To access applications from varied end-user devices (generally through a web browser)

Deployment Models

Private Cloud

Operated solely for one organization only or for an individual person

Public Cloud

Open to the general public or large industrial groups and are owned and are usually managed by a cloud service provider

Community Cloud

Enhanced security and privacy control. It can be located on-premises or off-premises.

Hybrid Cloud

Combining two or more clouds (private or public) that remain unique entities but are bound together with data and application portability.

Advantages of Cloud Computing

- Low costs
 - Outsourcing information reduces amount spent on new technology
 - Easier to maintain
- Interoperability
 - Access information from anywhere
 - Can be accessed using different devices



Advantages of Cloud Computing

- Increases the adoption of Electronic medical records.
- Beneficial for small companies
- Easy to share information among different organizations and doctors



Cloud Computing

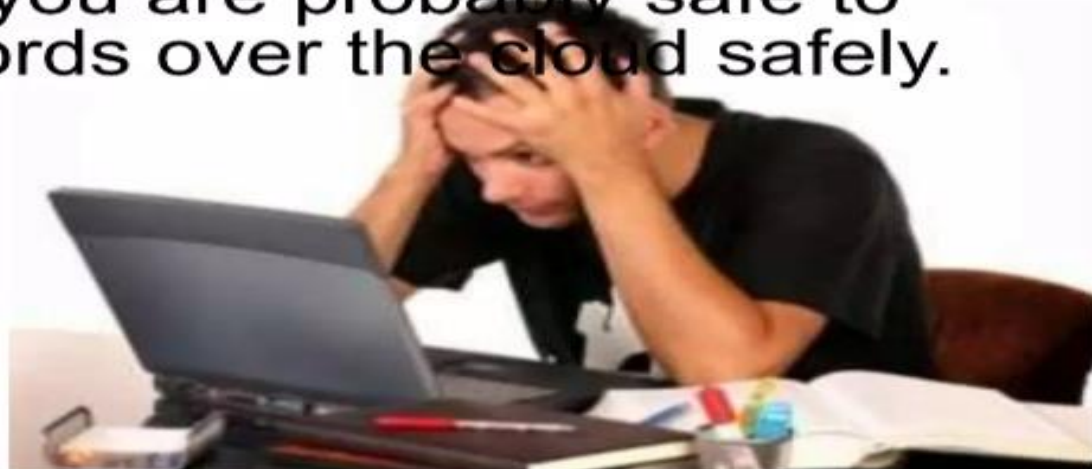
Disadvantages

- Security is the main disadvantage of cloud computing
- Consumers are worried about Insurance companies getting a hold of there information and discriminating based upon current medical conditions they may have or medical conditions that they could develop later in life.
- They are also worried about government agencies getting a hold of there information and exploiting it to third party vendors, or there place of work.



Disadvantages Cont.

- The cloud companies do not always handle all of the security themselves and sometimes pass it off to third party vendors
- Consumers need to make sure to thoroughly check out these companies to see who else they are involved with and check out there reputation to see if you trust them to not share your information with any other outside sources. If the third party vendor looks trustworthy then you are probably safe to send your medical records over the cloud safely.



Cloud Deployment in Healthcare

Data Centers



storage



Google Cloud Platform



Microsoft Azure



Patient Management

Billing

Radiology/Laboratory services

Electronic health record

Disaster Recovery

Remote location

Expensive

Identified Needs for Expansion of Cloud by Health Care

- **Cost Reduction**
 - The number one gain to be realized from the cloud is cost reduction. With all aspects, including infrastructure, software, and even management paid for on a per-use-basis, cloud greatly reduces the cost. No investment in any physical resources coupled with paying only for needed resources makes cloud an attractive option financially

Identified Needs for Expansion of Cloud by Health Care

- **Disaster Recovery**

The processes, policies and procedures to prepare for recovery or continuation of the organizations technology infrastructure after a natural or man-made disaster

Identified Needs for Expansion of Cloud by Health Care

- **Storage Scalability**

Maintaining storage and traffic load during the peak traffic load for a customer, without adding additional hardware or infrastructure without interference to the customer workload

Cloud service providers & Healthcare customers

SI No	Name of vendor	Healthcare Organization
1	CtrlS Data center, Hyderabad	Apollo Hospitals, Rainbow Hospitals, Call Health
2	Ricoh India data center	Fortis Hospital, Delhi /Artimis, Gurgaon / Emanuel Hospital
3	Microsoft Azure	L V Prasad Eye Institute, Hyderabad / Partners Health, United States / Apollo Hospitals / Fortis Hospitals
4	Amazon Cloud services	Cleveland Clinic, Ohio, United States



Conclusion

- Right Information at right time saves life.
- Economical and Efficient.
- Better management practice.
- Improve and solve several collaborative information issues in healthcare organizations.

THANK

YOU